



Continuous intensified selective extraction of platinum group metals in small channels

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ABSTRACT

Platinum-group metals (PGMs) are essential in various fields, including automotive emission control, process catalysis and emerging green energy technologies, making their efficient recovery crucial for sustainable industrial use. Solvent extraction (SX) is a key technique for the recovery and purification of these metals. This study explores the use of continuous small channel systems to intensify SX, focusing on the selective separation of palladium, platinum, and rhodium. By using two distinct extractants, trioctylamine, a fast outer-sphere ion-pair extractant, and di-n-hexyl sulfide, a slower coordination-based extractant, efficient, selective extraction was achieved. Trioctylamine achieved nearly 100 % extraction of palladium and platinum in just 7 s and 65 s respectively, whereas di-n-hexyl sulfide reached over 40 % palladium extraction within 5 min. In the small channel systems, mass transfer coefficients were found to be up to two orders of magnitude higher than in conventional SX equipment. Additionally, increasing the temperature from 20 °C to 40 °C led to a 50 % increase in mass transfer coefficients for the kinetically slow extraction with di-n-hexyl sulfide. These results suggest that small channel systems hold considerable promise for rapid, selective PGM recovery, supporting sequential multicomponent separation processes.

1. Introduction

Platinum-group metals (PGMs) are a subgroup of the transition metals comprising platinum, palladium, rhodium, ruthenium, iridium, and osmium. These metals hold great industrial significance due to their excellent catalytic properties, high melting points, and resistance to corrosion. Among them, platinum, palladium, and rhodium have been widely used in recent years driven by demand from the automotive sector in emission control applications. PGMs are also critical in other sectors, including large-scale industrial catalysis, electronics, and jewellery [1,2]. Recently, interest in the more niche PGM elements has grown, particularly due to their role in clean energy technologies and transition to net-zero; iridium, for example, is increasingly used in hydrogen production as a catalyst in proton exchange membrane (PEM) electrolyzers.

For many years PGMs were obtained from mining and ore processing in specific geographic areas (South Africa and Russia being the largest primary PGM suppliers), involving deep mining and energy-intensive, polluting processes with significant carbon footprints. As global demand and market value for PGMs rises, secondary refining and recycling

from catalysts and other PGM-containing products have become more viable alternatives. Established industrial methods for recycling platinum-group metals (PGMs) from spent catalysts involve both pyro- and hydrometallurgy [3]. Similarly to primary ore-based feed materials, PGM recycling feedstocks typically consist of complex mixtures of all the PGMs together, as well as many other associate base metals (such as copper and nickel) and associate elements (such as gold and silver). The high purity requirements for the PGM salt and metal products (typically 99.95 and 99.99 % respectively) place further demands on PGM recycling operations [4]. These challenges make developing more efficient and environmentally sustainable PGM recycling and recovery processes difficult.

In response to these challenges, solvent extraction (SX) has emerged as a robust technique for recovering and purifying PGMs obtained from diverse secondary recycling sources such as spent catalysts and electronic waste (e-waste) [5]. In SX, PGMs are separated from base and associate metals using functionalized organic extractants designed to exploit their chemical differences, offering advantages such as high selectivity, efficient metal recovery, and scalability [6]. PGM SX processes of this kind have already been successfully adopted at large

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commercial scale by major primary and secondary PGM producers and recyclers, including Anglo American Platinum, Johnson Matthey, and Vale INCO [7]. Despite this, opportunities remain to improve the sustainability of PGM SX processes and develop more flexible methods that can be adapted to different scales, locations and varying feed stocks from new technologies.

Recent research has focused on the challenges of separating PGMs like platinum (Pt) and palladium (Pd), which share similar physicochemical properties due to their position in the same periodic table group, by tailoring their extractant chemistry [8]. In industrial applications, PGM solvent extraction is most successfully carried out in aqueous chloride solutions where extractable PGM chloride complex anions of the general form $[MCl_x(H_2O)_y]^{n-}$ are the predominant species formed. The extractant chemistry can be adjusted to exploit this either *via* outer sphere ion pairing or inner sphere coordination type binding mechanisms to achieve selectivity. A number of inner sphere Pd coordinating or chelating extractants have been developed and commercially exploited to date including organosulfur [9], alkylated derivatives of 8-hydroxyquinoline [10] and hydroxy oximes [11]. Dialkyl sulfides and hydroxy oxime extractants have been reportedly used in large primary PGM refineries. These extractants are typically very selective towards palladium but they can suffer from slow kinetics which complicates efficient process design. To achieve multicomponent PGM SX separations, intricate multi-step processes are required using combinations of both inner and outer extractants together [7,11]. Such multistep processes must also include any associated pre-treatment procedures such as acidity and oxidation potential adjustment of the feed solutions as well as post treatment procedures like evaporation, filtration and metal recovery [12]. New SX process technologies that can simplify and combine steps together in a continuous flow mode would greatly improve the sustainability of PGM recycling *via* intensification.

Advances in SX equipment and process design, have improved phase mixing and mass transfer, ensuring that the extraction chemistry works at its optimum [13]. However, mass transfer kinetics still encounter hurdles in larger scale SX equipment. The slow inner sphere kinetic extraction mechanism is a particular challenge, and current mixer technology is often not well suited for achieving optimal mass transfer during extraction. Flow based solvent extraction within small channels has emerged as an intensified alternative to batch stirred tank and mixer settler SX processes, as it enhances mass and heat transfer rates, improving mixing and residence time control [14]. It also allows access to higher temperature regimes that would greatly accelerate the rate of inner sphere extraction mechanisms but are challenging to achieve in current SX equipment. The confined geometry and small scales give rise to new flow patterns for the two immiscible liquid phases used in SX [15–17]. The plug flow pattern observed in small channel systems, where one liquid forms elongated drops (plugs) within the continuous second liquid, has extensively been studied for SX processes [18]. Plug flow provides large specific interfacial areas, thin films, enhanced radial mixing via circulation patterns in both phases, and reduced axial mixing, thereby improving uniformity and intensifying mass transfer [19]. High extraction efficiencies greater than 90 % at low residence times have been reported in this regime for separations of biomolecules and of metals [20–22]. The improved mixing and thin fluid layers also enhance heat transfer, which is crucial for heat-sensitive extraction processes where maintaining optimal temperature conditions is crucial for product stability and selectivity.

Despite these advancements, flow extraction in small channels has predominantly focused on single-component separations, with limited studies on the combined effects of heat and mass transfer [17]. Additionally, the approach has not yet been demonstrated using established PGM SX extractants or tested for efficient sequential multicomponent PGM separations. The current work aims to address this gap by exploring the continuous selective extraction and separation of mixed palladium, platinum, and rhodium solutions in a small channel flow extraction system for the first time. By leveraging the process

intensification advantages of small channel flows, the work aims to enhance selectivity and extraction efficiency while addressing the time-sensitive selective extraction of palladium and platinum. Initially, experiments were conducted using high concentrations of PGMs to replicate real concentrations found in industrial leach solutions of spent automotive catalysts within commercial PGM recycling operations and demonstrate the limitations of the current extraction processes. Subsequent experiments in the small channel systems were performed using lower concentrations, given the limited availability of PGMs. For this, two well-known extractants, di-*n*-hexyl sulfide (DHS) and the strong base tri-*n*-octylamine (TOA) were chosen to represent the two extremes of slow (inner sphere mechanism) and fast (outer sphere mechanism) extraction kinetics. Extractions with both extractants were studied in small channels under the plug flow regime, from acidic solutions containing mixtures of two or three PGMs. In addition, the effect of temperature on the overall extraction percentage and mass transfer coefficients was investigated.

2. Experimental procedures

2.1. Materials

The stock solutions were prepared using sodium tetrachloropalladate (Na_2PdCl_4), chloroplatinic acid (H_2PtCl_6) and rhodium chloride ($RhCl_3$) obtained from Johnson Matthey (UK). Hydrochloric acid (HCl) of 37 wt % concentration, obtained from Sigma-Aldrich (UK), was used for adjusting the pH. Kerosene was obtained from Fisher Scientific (UK). The slow extractant, di-*n*-hexyl sulfide ($C_{12}H_{26}S$) at 97 % purity was purchased from Fisher Scientific (UK). The fast extractant, tri-*n*-octylamine ($C_{24}H_{51}N$) at a 98 % purity was purchased from Sigma-Aldrich (UK). All chemicals and solvents were used as received without further purification.

2.2. Apparatus and characterisation

The concentrations of the PGM feeds and aqueous extraction samples were measured with an Agilent 5110 inductively coupled plasma optical emission spectroscopic instrument (ICP-OES) using the 340.5 nm wavelength for palladium, 214.4 nm for platinum and 343.5 nm for rhodium. The kinetics of the small channel flow extractions were also monitored using Carey 60 UV-Vis spectrometer from Agilent. The UV-Vis measurements were recorded in 2 mm path length cuvettes using 3 M HCl as a reference. Acidities were determined by titration against a 1 M NaOH standard titrant solution using a Metrohm Titrando auto-titration system. Potentiometric end point determination was used with a glass working electrode (Radiometer pHG201-7) and a “Red-Rod” (Ag/AgCl in sat. KCl) reference electrode (Radiometer REF201) employing a continuous inflection point method to accurately calculate the end points. Oxidation potential was measured using a Pt wire working electrode and a “Red-Rod” (Ag/AgCl in sat. KCl) reference connected to the Metrohm Titrando system. Viscosities of the kerosene and aqueous phases were measured with a digital Rheometer DV-111 Ultra (Brookfield).

2.3. Sample preparation (Feed and extractant solution preparation)

For the batch experiments, an oxidised synthetic Rh/Pd/Pt feed solution was prepared with the composition reported in Table 1 (PGM solution 1). The concentrations used in the experiments replicate real concentrations found in industrial leach solutions from automotive catalytic converter recycling which contain palladium, platinum and rhodium. The concentrations used reflect the concentrations fed into the SX process following upstream treatment processes. High concentrations are more favourable for commercially viable solvent extraction processes.

For the synthetic aqueous feed preparation, the metal salts were

Table 1

Aqueous (3 M HCl) PGM feed stock for extraction experiments.

PGM solutions	Feed concentrations prepared (g/L)			Real concentrations (ICP) (g/L)		
	Palladium	Platinum	Rhodium	Palladium	Platinum	Rhodium
1	45	60	2.5	46	62	2.2
2	5	5	0.25	4.9	5.1	0.20
3	5	—	0.25	4.3	—	0.22
4	—	5	0.25	—	4.9	0.27

dissolved in concentrated HCl and made up to the target metal concentrations and 3 M acidity with de-ionised water. The solution was refluxed overnight with slow addition of drops of hydrogen peroxide to optimize the conditions for dissolving and oxidizing the metal salts. The oxidation potential of the solution was measured at 832.5 mV. The acidity was checked by titration of samples of the feed stock and adjustments made to achieve a final acidity of 3 M $\pm$ 0.1 M using appropriate amounts of water and concentrated hydrochloric acid. The organic phase for the batch DHS extraction kinetic experiment was prepared as a 25 % w/v solution of di-n-hexyl sulphide (DHS) in kerosene by adding 100 g to a 500 mL volumetric flask and making up to volume with the kerosene diluent.

Three further synthetic aqueous PGM feed solutions were prepared for the continuous small channel flow extractions with compositions reported in Table 1 (PGM solutions 2–4). The ratios of palladium to platinum and rhodium were the same as those in the high PGM concentration solution, but the concentrations were reduced by one order of magnitude to minimize costs associated with PGM materials and to explore how lower concentrations impact the efficiency of the extraction process.

The organic phase for both the DHS and TOA extractant experiments in the small channels was prepared using kerosene as the diluent. To ensure that the kerosene would not absorb water during the extraction process, it was first brought into contact with 3 M HCl. This step was essential to confirm that any subsequent metal extraction was solely attributed to the added extractants. Following this, the appropriate extractant was introduced into the kerosene. For the DHS extraction, a 33 % w/v solution of DHS was added to 100 mL of kerosene in a volumetric flask and the volume was made up to 500 mL. For the TOA extraction, an 11 % w/v solution of TOA in kerosene was prepared. The organic mixtures were stirred for one hour to ensure homogeneity prior to any extraction.

The combination of palladium, platinum, and rhodium in these three feeds allowed for the assessment of extraction from binary palladium/rhodium and platinum/rhodium mixtures as well as from three-component palladium/platinum/rhodium mixtures with each extractant. The inclusion of rhodium allowed testing of the feasibility of the complete sequential separation of palladium, platinum and rhodium using small channel flow extractors. Until recently, there have been very few reports of solvent extraction systems specifically for rhodium; typically, rhodium is left in the final raffinate after palladium and platinum extraction and is then recovered by other methods [23]. Rhodium which did not extract with either of the selected extractants, also served as a reference component in all three solutions to normalise and correct the ICP analysis results for any volume changes due to mutual phase solubility and any other losses during the extractions.

2.4. Extraction studies

2.4.1. Batch PGM DHS kinetic extraction experiments (High PGM concentrations)

A titanium stirred cell equipped with an overhead stirrer and a four bladed titanium impeller was immersed to $\frac{3}{4}$ of its height in a temperature-controlled water bath at 40 °C. The redox adjusted feed solution and the 25 % w/v DHS organic phase were immersed in the water bath and allowed to reach the required temperature. A portion of

the redox adjusted solution 1 (75 mL) was added to the stirred cell followed by a portion of the DHS organic phase (75 mL, phase ratio = 1:1) and stirring commenced (stirring speed = 1400 rpm). Samples of the aqueous phase were taken at 5, 10, 20, 40, 60, 90 and 120 min by stopping the mixing, collecting an aqueous and an organic sample, then restarting until the maximum time was reached. The aqueous samples were filtered with Whatman 540 paper to remove entrained organic phase and then analysed by ICP-OES.

2.4.2. Batch Pd DHS and Pt/Pd TOA equilibrium extraction experiments (Low PGM concentrations)

These experiments, using PGM feed solution 2 (Table 1), were designed to determine the maximum concentration of metal that could be extracted by the system under equilibrium conditions prior to the continuous small channel system extractions. The data obtained from these equilibrium tests were then used to calculate the percentage extraction and the extraction efficiency of the small channel system. In these experiments, 5 mL of the organic kerosene phase, containing one of the two extractants (DHS or TOA), was mixed with 5 mL of the aqueous metal feed stock solution. The mixture was continuously stirred in a temperature-controlled water bath set to either 20 °C or 40 °C. The contact time between the two phases varied, in intervals of 1 min, 2 min, 5 min, 30 min, 60 min, and 120 min. The phases were separated and the aqueous phases after each time interval were analysed in the ICP OES and UVVis as outlined in section 2.2.

2.4.3. Experimental set-up for flow extraction

Experiments on continuous flow extraction in small channels were conducted using the set-up illustrated in Fig. 1. Kd Scientific high precision syringe pumps were used to feed the organic and the aqueous phases separately in the main channel made of PTFE via a T-junction inlet configuration made of FEP. All feeding channels had the same internal diameters as that of the main one (0.5 mm). The organic phase was introduced from the side channel at the same axis as the main one and acted as the continuous phase in contact with the channel wall, while the aqueous phase was introduced from the side channel perpendicular to the main one. The extraction experiments were performed in channels with lengths that varied from 15 to 390 cm, to control the residence time. The flow patterns in the main channel were imaged using a Photron Fastcam-ultima high speed camera at frame rates between 2000 to 4000 Hz, depending on the flowrate. At the end of the channel test section, the two phases were separated in-line using a membrane-based flow separator (Zaiput Flow Technologies, SEP-10). The membrane is hydrophobic and made from PTFE, thus allowing only the kerosene phase to permeate it, while the aqueous phase passed over it. The concentrations of the metals in the aqueous phase after the separator were measured with the ICP OES and UV-Vis. The metal concentration in the kerosene phase was determined from the mass balance between the aqueous phase concentration in the outlet and in the initial feed, assuming no phase volume changes.

In the experiments carried out at 40 °C, the channels before the T-junction were immersed in a heated bath regulated by a temperature loop, to ensure that the solution was at 40 °C before entering the T-junction as depicted in Fig. 1. Similarly, the main channel was submerged in the water bath with temperature maintained at 40 °C.

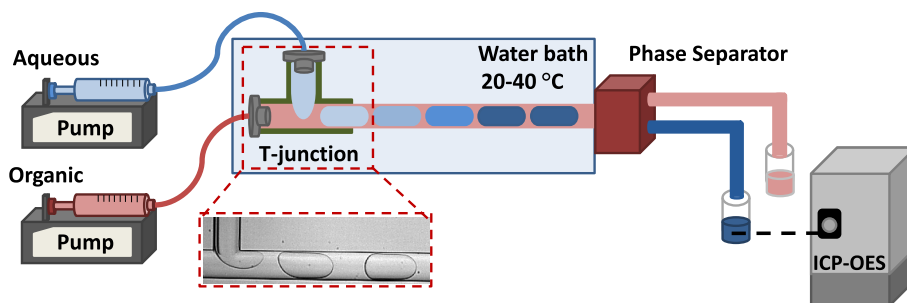


Fig. 1. Experimental set-up of small channel system showing continuous flow extraction of PGMs.

3. Extraction and separation mechanisms

Aqueous chloride solutions are the standard media for PGM extraction due to formation of stable PGM chloride complexes, which facilitate selective separation. The selective separation of rhodium, palladium and platinum depends on the relative charge, size and shape of their respective chloro anion complexes and the stability of their metal chloride ligand coordination bonds [24]. For example, both the 2+ and 4+ oxidation states of palladium and platinum are known but Pd^{4+} is very uncommon as it is only stable at high equilibrium redox potentials (>1.29 V) for the transformation of $[\text{PdCl}_4]^{2-}$ to $[\text{PdCl}_6]^{2-}$. In contrast, Pt has an equilibrium redox potential of 0.74 V for the transformation of $[\text{PtCl}_4]^{2-}$ to $[\text{PtCl}_6]^{2-}$ and so can adopt both oxidation states [12]. The principal extractable chloro anions of Pd and Pt are illustrated in Fig. 2.

When PGMs dissolve and leach from spent catalysts, the oxidation potential of the solutions is too high for Pt^{2+} to form, and so Pt^{4+} is the only oxidation state present at equilibrium. Feed solutions in this study were thus prepared by refluxing overnight in air with addition of some hydrogen peroxide to ensure the exclusive presence of the $[\text{PtCl}_6]^{2-}$ over $[\text{PtCl}_4]^{2-}$ chloro anions. Rhodium only exists in the 3+ oxidation state under these conditions. At 3 M HCl, rhodium forms a series of aquated chloride species $[\text{RhCl}_{6-n}(\text{H}_2\text{O})_n]^{3-(6-n)}$ that do not extract with either DHS or TOA under these conditions. The palladium, platinum, and rhodium chloride complexes can undergo stepwise aquation, forming various aquated chloride species depending on metal and chloride concentrations. The different complexes of each element within the same solution can affect their extraction efficiency. For this reason, high chloride concentrations are generally used to ensure that only the fully chlorinated Pd and Pt anions are present. Therefore, all the PGM feed solutions were adjusted to 3 M HCl concentration for all the experiments.

The use of tertiary amines ($\text{R}_3\text{NH}^+\text{Cl}^-$) as outer sphere extractants for Pd and Pt chloride anions has been previously reported (Matthey-Rustenberg Refiners process) [25]. Here, the tertiary amine, TOA, is chosen to demonstrate the application of small channel flow extraction to the fast outer sphere extraction mechanism. As a strong base, TOA is fully protonated at all pH values and the extraction is governed by the anion exchange equilibria between TOAH^+Cl^- and the palladium and platinum chloro anions as given in Eqs. (1) and (2).

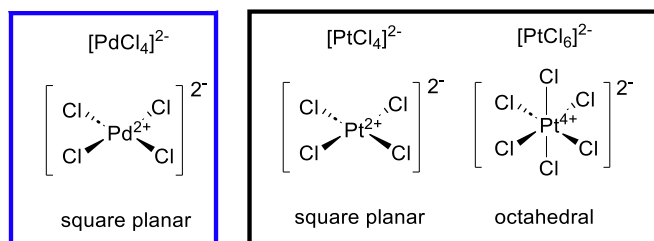
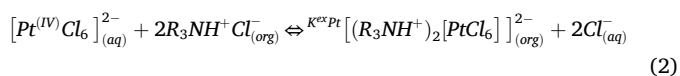
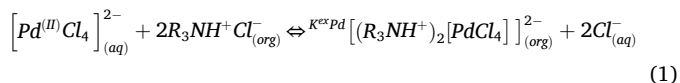
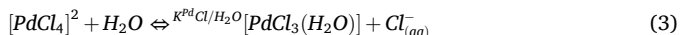


Fig. 2. Oxidation state and structural configuration of the most common Pd and Pt chloro-complexes.

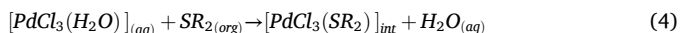


Both the $[\text{PtCl}_6]^{2-}$ and the $[\text{PdCl}_4]^{2-}$ anions are strongly extracted by TOA. In contrast, the rhodium complex, $[\text{RhCl}_6]^{3-}$, does not form strong extractable ion pairs with the $(\text{TOA})\text{H}^+$ cation and is not extracted. The aquated rhodium chloride anions, $[\text{RhCl}_5(\text{H}_2\text{O})]^{2-}$ and $[\text{RhCl}_4(\text{H}_2\text{O})_2]^-$ present at 3 M HCl concentration, also do not extract due to their significant hydration energies which makes phase transfer into nonpolar organic solvents unfavourable, consistent with the Hofmeister series [23].

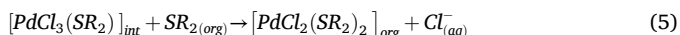
DHS is an established inner sphere palladium extractant [26]. It was therefore chosen as a good example to demonstrate this Pd solvent extraction mechanism in intensified small channel systems. As Pd^{2+} ion is significantly more labile than either the Rh^{3+} or Pt^{4+} , its chloro complex $[\text{PdCl}_4]^{2-}$ undergoes more rapid chloride water exchange (Eq. (3)), with a reaction half-life of less than 0.1 s, making it liable to further ligand substitution [8,25].



In contrast, $[\text{PtCl}_6]^{2-}$ is inert to ligand substitution and far less liable to chloride water exchange. The substitution rate for the square planar aquated $[\text{PdCl}_3(\text{H}_2\text{O})]^-$ complex is several orders of magnitude greater than for Pt or Rh making it far more susceptible to coordination by strong donor extractant ligands like DHS Eq. (4).



Consequently, DHS is highly selective in extracting Pd. The intermediate $[\text{PdCl}_3(\text{SR}_2)]^-$ complex that forms at the organic/aqueous interface undergoes further chloride ligand exchange at the interface to form the extracted $[\text{PdCl}_2(\text{SR}_2)_2]$ complex within the organic phase Eq. (5).



This second step has been shown to be rate limiting [27] and the overall extraction rate for Pd with DHS is therefore dependent on the interfacial area. Poor mixing and low interfacial areas will limit the observed overall mass transfer rate. In batch stirred tanks or mixer-settler SX equipment, efficient mixing requires energy input from an impeller. This can lead to poor droplet size control and non-optimum interfacial mass transfer, whereas in the small channel flow extraction systems the droplet size and mass transfer rate can be well controlled by the flow regime.

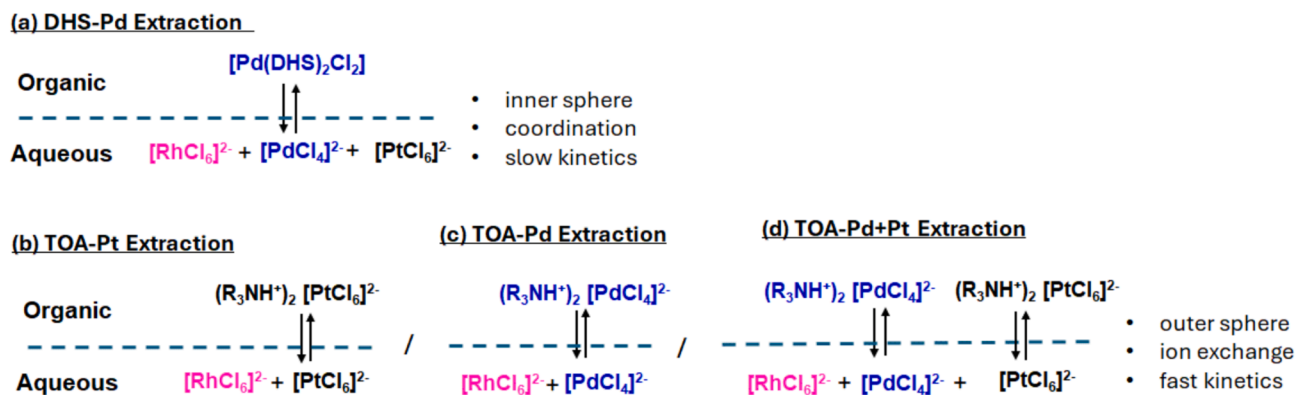


Fig. 3. Extraction of palladium and platinum chloro-complexes with TOA and DHS showing: (a) Selective Pd extraction with DHS from Rh and/or Pt, (b), (c) and (d) anion exchange SX of Pd, Pt, and Pd&Pt from Rh.

4. PGM solvent extraction schemes

The application of the small channel flow solvent extraction system enables a variety of separation schemes. The combination of the fast and slow extraction mechanisms allows the four main SX separation schemes shown in Fig. 3. DHS on its own allows selective extraction of palladium from either binary mixtures of Rh/Pd and Pt/Pd or from ternary mixtures of Rh/Pd/Pt (Fig. 3 a). On the other hand, TOA on its own allows extraction and separation of either Pt and/or Pd from rhodium but not selective separation of Pd from Pt (Fig. 3 b,c and d). The combination of DHS and TOA in sequence allows for the complete selective separation of Rh, Pd and Pt in small channel flows.

5. Batch DHS extraction of Pd from a high concentration mixed PGM feed

Conventional batch extraction of palladium from mixed PGM solutions using di-*n*-alkyl sulphides is well established [28] but challenges have been reported that limit their utilization. In addition to the slow kinetics at room temperature, the co-extraction of other PGM elements has been reported, despite the theoretical selectivity of DHS for Pd over other PGMs via the coordination mechanism. This co-extraction is reported to be related to the kinetics of the extraction and competing side reactions [29].

To test the effect of kinetics on selectivity, a batch palladium DHS extraction was performed (see section 2.4.1) from the mixed PGM feed solution 1 (Table 1) in a stirred cell system. This feed contained realistic industrial process concentrations of rhodium, palladium, and platinum that have been adjusted to a higher oxidation potential of 832.5 mV to ensure Pt^{2+} was not present.¹ This solution was contacted with an organic phase containing the DHS extractant in the stirred cell at 40 °C and at high mixing rate (1400 rpm) for 60 min. Higher temperature was applied to increase the rate of reaction to a practical time as previous data¹ showed that at room temperature the extraction was very slow.

Fig. 4 demonstrates that achieving 100 % palladium extraction using the conventional batch approach is inherently challenging, as some platinum co-extraction occurs before complete palladium recovery. This observation aligns with findings reported in the literature [27]. As expected, the extraction kinetics of palladium were faster at 40 °C than at room temperature but nevertheless 20 min were still required to achieve 100 % extraction. After 5 min 90 % Pd extraction occurred with no co-extraction of either platinum or rhodium. However, after 10 min when 98 % Pd extraction was achieved co-extraction of the Pt was observed, between 0.1 and 0.2 %. A small amount of rhodium extraction was also observed. When complete palladium extraction is achieved at 20 min

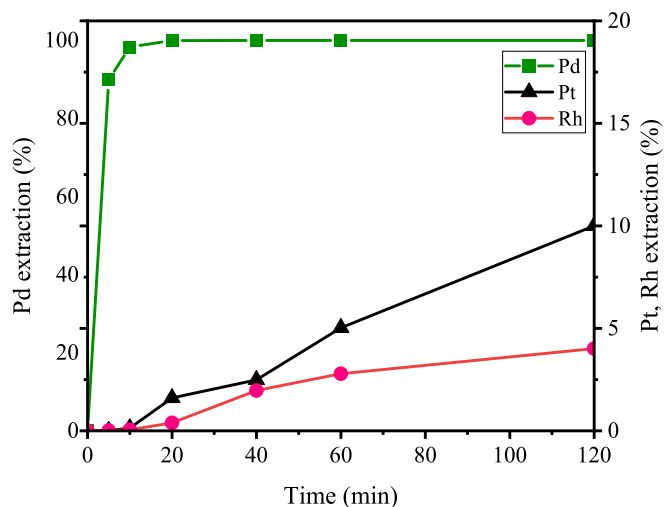


Fig. 4. Batch extraction of oxidised Rh/Pd/Pt feed solution 1 using DHS extractant at 40 °C .

significant co-extraction of platinum (1.6 %) is observed. This would lead to reduced purity of any recovered palladium in conventional stirred contacting equipment where accurate control of the residence time is difficult and there is therefore, a trade-off between recovery yield and purity. The time scales here for the extraction of Pd, Rh, and Pt agree with extraction of PGMs performed in conventional lab scale mixer-settlers units. The extraction times observed for Pd, Rh, and Pt in this study are consistent with those reported for extraction of platinum group metals in conventional lab-scale mixer-settler units [30].

It has been reported that this co-extraction occurs because of the oxidation of the DHS extractant to di-*n*-hexyl sulfoxide (DHSO) [29]. The DHSO produced can act as an outer sphere extractant and co-extracts other PGMs anions such as $[\text{PtCl}_6]^{2-}$ by ion pair binding. This oxidation is reported to be catalysed by the Pt^{4+} in the feed and results in co-reduction of Pt^{4+} to Pt^{2+} which forms the extractable Pt complex, $\text{PtCl}_2(\text{DHS})_2$. In practical applications, the oxidation of DHS to DHSO would also reduce the lifetime of the extractant. This issue is particularly pronounced in conventional solvent extraction units, where temperature control and mixing are often not uniform. On the other hand, small channel extraction enables precise control of mixing, temperature, and residence time as well as efficient separation of the two phases at the end.

¹ Information supplied by Johnson Matthey.

6. Continuous flow extraction

To overcome the issues of batch separations stated above, small channels are used to enhance the efficiency of the extraction process. In the experiments that follow, lower PGM concentrations are used (solutions 2, 3, and 4 from Table 1) due to limited availability of PGMs. For the extractions in small channels, the plug flow pattern was chosen, which offers enhanced mass and heat transfer rates. In this regime, the continuous phase comprises the TOA or DHS extractants in kerosene, while the aqueous phase forms convex-shaped plugs within it. Different phase flowrates and volume fractions were tested to obtain the conditions for plug flow and a flow pattern map, depicted in Fig. 5, was composed.

For all the extraction experiments an overall flowrate equivalent to a mixture velocity of 0.01 m/s, was chosen at equal flowrates of the two phases (or volume fraction of 0.5). The extractant to palladium ratio was 1:50 for all experiments determined from initial equilibrium experiments.

To characterise the mass transfer performance of the flow extractors, the percentage extraction, P%, and the overall volumetric mass transfer coefficient, $k_L\alpha$, were calculated as follows [31].

$$P\% = \frac{C_{aq,in} - C_{aq,fin}}{C_{aq,in}} * 100 \quad (6)$$

$$k_L\alpha = \frac{1}{\tau} \ln \left(\frac{C_{aq,eq} - C_{aq,fin}}{C_{aq,eq} - C_{aq,in}} \right) \quad (7)$$

where $C_{aq,in}$ and $C_{aq,fin}$ are the initial and final aqueous phase concentrations of the PGMs, respectively. $C_{aq,eq}$ is the maximum concentration that can be extracted determined during the equilibrium experiments. The residence time, τ (s) of the plugs in the channel is calculated as $\tau = V_{ch}/Q_{tot}$, where V_{ch} is the channel volume and Q_{tot} is the total volumetric flow rate.

The following experiments were conducted in small channels to evaluate the extraction efficiency of palladium and platinum. Initially, a metal solution containing Pd, Pt, and Rh was contacted with the organic phase containing the slow extractant DHS, which selectively extracts only palladium. Given the temperature sensitivity of this extraction, the effect of elevated temperatures on palladium extraction was

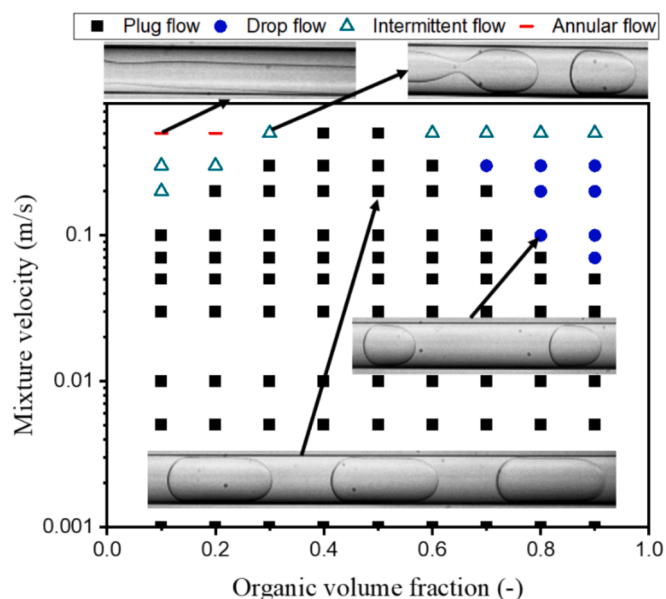


Fig. 5. Flow pattern map for kerosene-aqueous system in a 0.5 mm channel internal diameter.

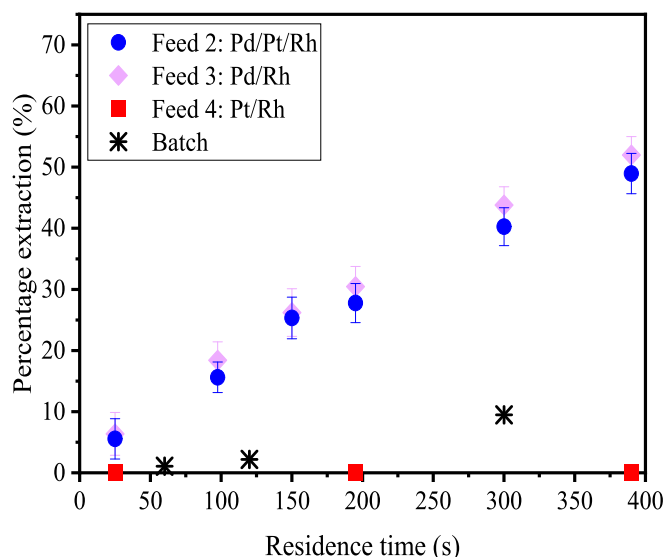


Fig. 6. Percentage extraction at 20 °C of palladium in small channels versus batch experiments using DHS.

subsequently investigated. In the next set of experiments, the fast extractant TOA, capable of extracting both palladium and platinum, was employed to assess its impact on the extraction of both metals. Rhodium, which does not extract with either extractant, served as a control component throughout the experiments, providing a consistent baseline for comparison. Rhodium was also included to demonstrate the capability for complete inter-separation of palladium, platinum and rhodium from a mixed feed solution. The continuous flow experiments were conducted at low metal concentration (5 g/L Pd and Pt, 0.2 g/L Rh) in the feed (PGM solutions 2–4 in Table 1).

6.1. Continuous flow extraction with low concentration PGM feeds- DHS

In these experiments, the DHS extractant facilitated the selective extraction of palladium from platinum and rhodium. Extractions were performed both in batch experiments to obtain equilibrium concentrations with metal feed 2 containing all three PGMs, and in the small channel flow system with feeds 2, 3 and 4 found in Table 1.

As can be seen from Fig. 6, a palladium percentage extraction (calculated from Eq. (6)) of over 40 % can be achieved within 5 min in the small channel with both feeds. The slightly higher extraction percentage observed with feed 3 that only had two metals compared to feed 2, is attributed to the slightly higher initial concentration of palladium in solution 2 at 4.9 g/L compared to stock solution 3 at 4.3 g/L. In comparison, less than 10 % extraction was obtained in 5 min in the batch experiments under similar conditions. This difference highlights the enhanced mass transfer capabilities of the small channel system, even when using a kinetically limiting extractant. Additional flow experiments conducted using metal feed 4 (containing only platinum and rhodium) showed no detectable extraction of Pt or Rh, within the tested residence times, confirming the selectivity of DHS for palladium in the small channel system.

6.1.1. Effect of temperature on continuous DHS extraction

To assess the impact of temperature on the flow extraction (as discussed in Section 5) with the DHS extractant, feed 3 was used, which contains palladium and rhodium. Figs. 7 and 8 present the impact of temperature on both the percentage extraction and the overall mass transfer coefficient ($k_L\alpha$), respectively, for the palladium separation in small channels. As it can be seen, no rhodium was detected in the organic phase at both temperatures, indicating the high selectivity of the extractant for palladium. A 50 % percentage extraction was achieved

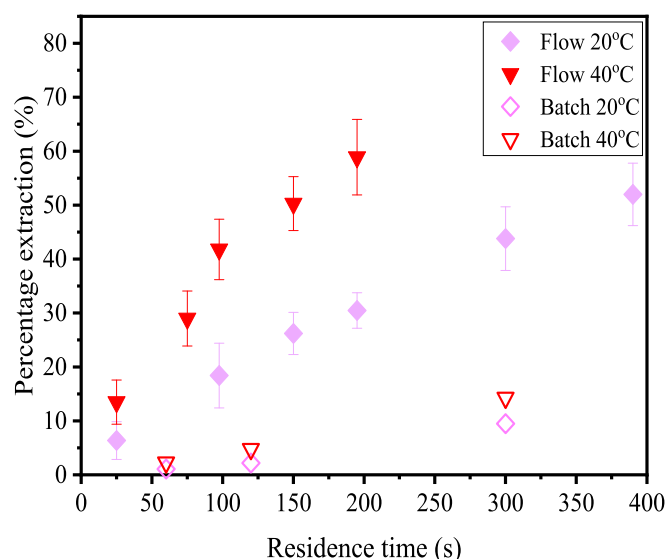


Fig. 7. Feed solution 3 percentage extraction of palladium in the presence of rhodium in small channels for 20 °C and 40 °C.

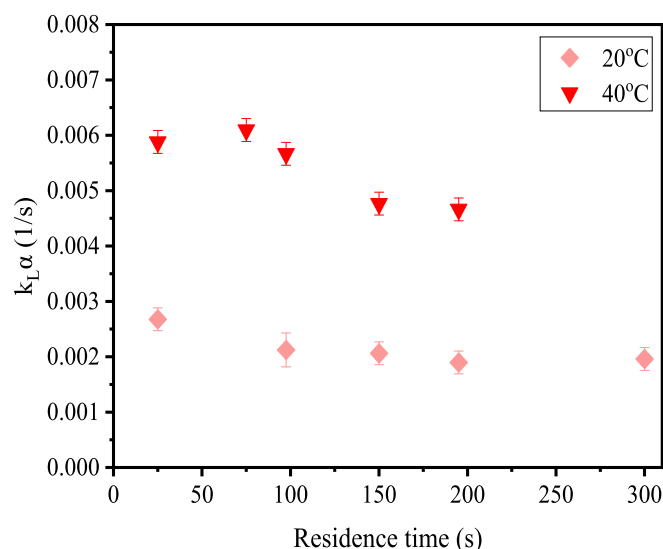


Fig. 8. Feed solution 3 overall volumetric mass transfer coefficient of palladium in the presence of rhodium in small channels for 20 °C and 40 °C.

within 2.5 min at 40°C, compared to approximately 5 min required at ambient temperature.

Similarly, the overall mass transfer coefficients were increased at the elevated temperature, with values at 40 °C about twice those observed at 20 °C (Fig. 8). Additionally, the $k_L\alpha$ values were an order of magnitude higher in the channels than in batch, where they were calculated as $6.0 \times 10^{-4} \text{ s}^{-1}$ at 40°C after 2.5 minutes. The higher $k_L\alpha$ values observed at elevated temperature reflect the increase in the reaction rate (see Eq. (7)) and improved mass transfer due to the decrease in viscosity of the fluids. Kerosene viscosity decreases from 1.8 mPa.s at 20 °C to 1.3 mPa.s at 40 °C while the aqueous phase viscosity decreases from 1.1 mPa.s to 0.7 mPa.s. Low viscosities enhance the circulation patterns and circulation velocity within the two phases resulting in more intense mixing [17,32].

For both temperatures, larger $k_L\alpha$ values are found at short residence times. This is attributed to the larger concentration gradients at the inlet of the channels which drive mass transfer. In addition, mixing is more intense at the T-junction where the plugs form. As the residence time

increases, the influence of the inlet conditions becomes less pronounced and the overall value of $k_L\alpha$ decreases and approaches a constant. This change in $k_L\alpha$ due the inlet effects is more pronounced at 40°C compared to 20°C. The better heat transfer properties and temperature control in small channel flow systems, demonstrated here, offer the potential to use heat transfer efficiently and reduce extraction times with inner sphere ligands, something that is hard to achieve in conventional extraction systems at process scale.

6.2. Continuous flow extraction with low concentration PGM feeds – TOA

Fig. 9 illustrates the percentage extraction of palladium, platinum, and rhodium with the fast extractant TOA at room temperature using feed solutions 3 and 4. The data confirm the ability of the extractant to selectively extract palladium and platinum from rhodium. As it can be seen, palladium is completely extracted within just 7 s while Pt requires 65 s. These findings are consistent with prior literature reports detailing similar extraction times for platinum and palladium observed in small channels operating in parallel flow using comparable amine-based extractants [33].

Despite the overall fast extraction rates, a difference in the extraction times of palladium and platinum can be seen. Palladium is extracted faster than platinum by approximately 60 s. This is likely due to the difference in size and charge density between the square planar $[\text{PdCl}_4]^{2-}$ dianion and the more symmetric octahedral $[\text{PtCl}_6]^{2-}$. The precise mechanism, however, is not known and warrants further investigation. In practice, this makes palladium more readily available for extraction by fast amine-based extractants. In conventional mixer-settler units, the selective separation of palladium and platinum using the fast extractant TOA is not possible, because of lack of precise control over the residence time [34]. The findings here, suggest that it is possible to selectively extract Pd from Pt even with the fast ion-pairing extractants by controlling the residence times.

The overall mass transfer coefficients $k_L\alpha$ for palladium and platinum using the fast TOA extractant are shown in Fig. 10. The results indicate that the $k_L\alpha$ values for palladium are consistently higher than those for platinum; they are about double at long residence times, while at short residence times below 20 s they are almost 5 times greater. This highlights the significantly faster extraction kinetics for palladium at the initial stages of the process, where a large concentration gradient drives rapid mass transfer. For both metals, increasing residence time results in

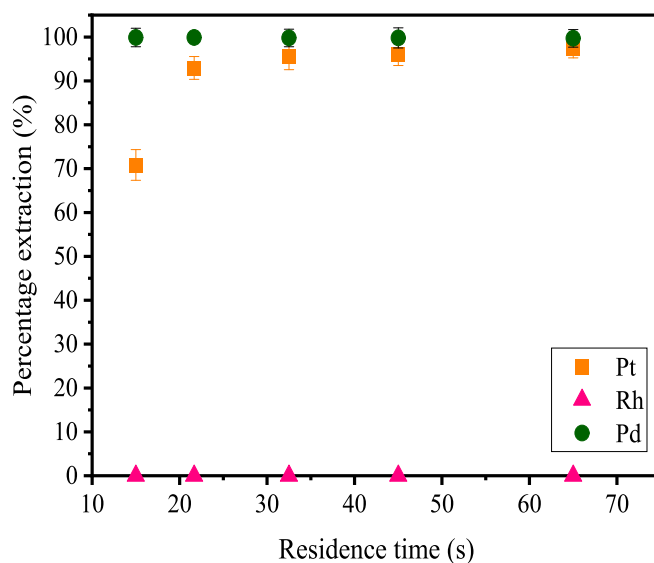


Fig. 9. Percentage extraction of platinum (Pt), palladium (Pd) and rhodium (Rh) using trioctylamine at 20 °C.

a general decline in $k_L\alpha$ which can be attributed to the decreasing concentration gradient between the phases as extraction progresses. Interestingly, platinum shows a slight increase in $k_L\alpha$ around 20 s, compared to 15 s, where close to 100 % extraction has been achieved in 20 s. As residence time increases further, $k_L\alpha$ continues to decrease, indicating that additional residence time beyond the near completion of the extraction offers minimal improvement in mass transfer. For palladium, which rapidly achieves near-complete extraction, the decrease in $k_L\alpha$ is even more pronounced. Furthermore, the $k_L\alpha$ values obtained with the fast extractant (Fig. 10) are two orders of magnitude greater than those of the slow extractant (Fig. 8) that were in the range of 10^{-3} s^{-1} . These values are also one order of magnitude greater than those observed in conventional units, such as mixer-settlers [35].

Based on the results, a process can be proposed where palladium is first separated from a mixed feed with platinum and rhodium using DHS in a continuous small channel flow system. In the next step, platinum is separated from rhodium using TOA. These two steps can be directly linked, with the raffinate from the DHS extraction fed into the TOA extraction. The system can be implemented in multistage configurations to enhance separation efficiency.

7. Conclusions

This study demonstrates the effectiveness of small channel solvent extraction systems for the selective separation of platinum, palladium and rhodium. Compared to conventional batch methods, these systems significantly enhance mass transfer rates and extraction efficiency.

Two different types of extractants were tested: a fast outer-sphere extractant, tri-n-octylamine (TOA), and a slow inner-sphere extractant, di-n-hexyl sulphide (DHS). Using the DHS extractant, selective palladium extraction was achieved with over 40 % extraction in 5 min at 20 °C, improving to 50 % extraction in 2.5 min at 40 °C with no platinum or rhodium co-extraction. Additionally, after 2.5 min in the small channel system, the overall mass transfer coefficient at 40 °C was approximately $5.0 \times 10^{-3} \text{ s}^{-1}$ compared to $6.0 \times 10^{-4} \text{ s}^{-1}$ in the batch system.

With TOA both palladium and platinum chloro-complexes were rapidly extracted, achieving 100 % palladium extraction in 7 s and platinum in 65 s. In addition, in the small channel system it was found that the palladium chloro anions extracted significantly faster than the platinum ones. This is the first time this has been observed, as such short residence times and accurate control are not achievable in conventional batch units. TOA demonstrated overall mass transfer coefficients 2 orders of magnitude greater than those of DHS and 1–2 orders of magnitude greater than those achieved in conventional solvent extraction units.

The results demonstrate that a complete, selective and continuous SX separation of palladium, platinum, and rhodium is possible using small channel flow extraction systems. A two-step process is proposed where DHS at > 40 °C extracts palladium first, followed by TOA, which extracts platinum while rhodium remains in the aqueous phase. These findings highlight the potential of small channel SX systems for PGM recovery from industrial feeds.

CRediT authorship contribution statement

Yiota-Victoria Phakoukaki: Conceptualization, Funding acquisition, Investigation, Methodology, Writing – original draft, Formal analysis. **Paul O'Shaughnessy:** Supervision, Conceptualization, Writing – review & editing, Validation. **Panagiota Angeli:** Supervision, Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: [Yiota Victoria Phakoukaki reports financial support was provided by

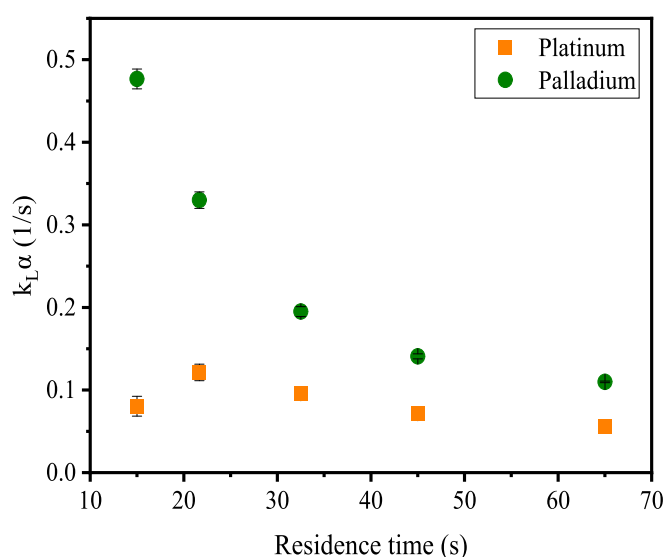


Fig. 10. Overall mass transfer coefficient of Pt and Pd using trioctylamine.

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Data availability

Data will be made available on request.

References

- [1] A.E. Hughes, N. Haque, S.A. Northey, S. Giddey, Platinum group metals: a review of resources, production and usage with a focus on catalysts, *Resources* 10 (9) (2021) 93, <https://doi.org/10.3390/resources1009093>.
- [2] Johnson Matthey. (2024). PGM market report. In PGM market report (Issue May).
- [3] C. Saguru, S. Ndlovu, D. Moropeng, A review of recent studies into hydrometallurgical methods for recovering PGMs from used catalytic converters, *Hydrometall.* 182 (2018) 44–56, <https://doi.org/10.1016/j.hydromet.2018.10.012>.
- [4] Anderson, C. G., & Cui, H. (Eds.). (2021). Advances in Mineral Processing and Hydrometallurgy. *Metals* 11(9), 1393. <https://doi.org/10.3390/met11091393>.
- [5] Y. Ding, S. Zhang, B. Liu, H. Zheng, C. Chang, C. Ekberg, Recovery of precious metals from electronic waste and spent catalysts: a review, *Resour. Conserv. Recycl.* 141 (2019) 284–298, <https://doi.org/10.1016/j.resconrec.2018.10.041>.
- [6] J.Y. Lee, B. Raju, B.N. Kumar, J.R. Kumar, H.K. Park, B.R. Reddy, Solvent extraction separation and recovery of palladium and platinum from chloride leach liquors of spent automobile catalyst, *Sep. Purif. Technol.* 73 (2) (2010) 213–218, <https://doi.org/10.1016/j.seppur.2010.04.003>.
- [7] Crundwell, F.K., Moats, M.S., Ramachandran, V., Robinson, T.G., Davenport, W.G. (2011). Extractive Metallurgy of Nickel, Cobalt and Platinum Group Metals. Elsevier. <https://doi.org/10.1016/C2009-0-63541-8>.
- [8] A.P. Paiva, Recent advances in platinum and palladium solvent extraction from real leaching solutions of spent catalysts, *Catalysts* 13 (7) (2023) 1146, <https://doi.org/10.3390/catal13071146>.
- [9] J.E. Barnes, J.D. Edwards, Solvent extraction at Inco's Acton precious metal refinery, *Chem. Ind.* 6 (5) (1982) 151.
- [10] G.P. Demopoulos, Solvent extraction in platinum and palladium refining, *JOM* 38 (6) (1986) 13–17, <https://doi.org/10.1007/BF03257809>.

- [11] M.V. Rane, PGM ore processing: LIX reagents for palladium extraction & platinum stripping from Alamine 336 using NaOH-NaCl, *Miner. Eng.* 138 (2019) 119–124, <https://doi.org/10.1016/j.mineng.2019.04.044>.
- [12] F.L. Bernardis, R.A. Grant, D.C. Sherrington, A review of methods of separation of the platinum-group metals through their chloro-complexes, *React. Funct. Polym.* 65 (3) (2005) 205–217, <https://doi.org/10.1016/j.reactfunctpolym.2005.05.011>.
- [13] J. Wang, W. Xu, H. Liu, F. Yu, H. Wang, Extractant structures and their performance for palladium extraction and separation from chloride media: a review, *Miner. Eng.* 163 (2021) 106798, <https://doi.org/10.1016/j.mineng.2021.106798>.
- [14] A. Mitic, K.V. Gernaey, Process intensification tools in the small-scale pharmaceutical manufacturing of small molecules, *Chem. Eng. Technol.* 38 (10) (2015), <https://doi.org/10.1002/ceat.201400765>.
- [15] J. Jovanović, W. Zhou, E.V. Rebrov, T.A. Nijhuis, V. Hessel, J.C. Schouten, Liquid–liquid slug flow: hydrodynamics and pressure drop, *Chem. Eng. Sci.* 66 (1) (2011), <https://doi.org/10.1016/j.ces.2010.09.040>.
- [16] D. Tsoulidis, V. Dore, P. Angeli, N.V. Plechkova, K.R. Seddon, Flow patterns and pressure drop of ionic liquid–water two-phase flows in microchannels, *Int. J. Multiph. Flow* 54 (2013) 1–10, <https://doi.org/10.1016/j.ijmultiphaseflow.2013.02.002>.
- [17] Q. Zhang, H. Liu, S. Zhao, C. Yao, G. Chen, Hydrodynamics and mass transfer characteristics of liquid–liquid slug flow in microchannels: the effects of temperature, fluid properties and channel size, *Chem. Eng. J.* 358 (2019) 794–805, <https://doi.org/10.1016/j.cej.2018.10.056>.
- [18] Y.-V. Phakoukaki, P. O'Shaughnessy, P. Angeli, Flow patterns of ionic liquid based aqueous biphasic systems in small channels, *Chem. Eng. Sci.* 265 (2023) 118197, <https://doi.org/10.1016/j.ces.2022.118197>.
- [19] C.X. Zhao, A.P.J. Middelberg, Two-phase microfluidic flows, *Chem. Eng. Sci.* 66 (7) (2011), <https://doi.org/10.1016/j.ces.2010.08.038>.
- [20] N. Assmann, A. Ładosz, P. Rudolf von Rohr, Continuous micro liquid–liquid extraction, *Chem. Eng. Technol.* 36 (6) (2013) 907–914, <https://doi.org/10.1002/ceat.201200557>.
- [21] Y.-V. Phakoukaki, P. O'Shaughnessy, P. Angeli, Continuous plug flow extraction of L-tryptophan using ionic liquid-based aqueous biphasic systems in small channels, *Sep. Purif. Technol.* 125468 (2023), <https://doi.org/10.1016/j.seppur.2023.125468>.
- [22] C. Pheasey, P. Angeli, Intensified Nd extraction in small channels for NdFeB magnet recycling, *Sep. Purif. Technol.* 311 (2023) 122958, <https://doi.org/10.1016/j.seppur.2022.122958>.
- [23] A.I. Carrick, J. Patrick, E.R. Schofield, P. O'Shaughnessy, B. Breeze, J.B. Love, C. A. Morrison, Separation of rhodium from iridium through synergistic solvent extraction, *Sep. Purif. Technol.* 333 (2024) 125893, <https://doi.org/10.1016/j.seppur.2023.125893>.
- [24] M.J. Cleare, P. Charlesworth, D.J. Bryson, Solvent extraction in platinum group metal processing, *J. Chem. Technol. Biotechnol.* 29 (4) (1979) 210–224, <https://doi.org/10.1002/jctb.503290403>.
- [25] J. Rydberg (Ed.), *Solvent extraction principles and practice*, revised and expanded, CRC Press, 2004, <https://doi.org/10.1201/9780203021460>.
- [26] S. Daamach, G. Cote, D. Bauer, Separation of platinum group metals in hydrochloric media: solvent extraction of palladium (II) with dialkyl sulphides, *Sep. Process. Hydrometallurgy* 304 (15) (1987) 889–892.
- [27] Y. Baba, T. Eguchi, K. Inoue, Solvent extraction of palladium with dihexyl sulfide, *J. Chem. Eng. Jpn* 19 (5) (1986) 361–366, <https://doi.org/10.1252/jcej.19.361>.
- [28] Edwards, R.I. (1979). Selective solvent extractants for the refining of platinum metals. In B.H. Lucas, G.M. Ritecy, & H.W. Smith (Eds.), *Proceedings of the International Solvent Extraction Conference (ISEC '77)*, Toronto, 9–16 September 1977 (Vol. 1, pp. 24–30). Canadian Institute of Mining and Metallurgy.
- [29] A. Okuda, H. Sawai, S. Ichiishi, J. Shibata, Extractions of Pd(II) and Pt(IV) from hydrochloric acid solution by dihexyl sulfide and the degradation of dihexyl sulfide, *Shigen-to-Sozai* 116 (11) (2000) 929–933, <https://doi.org/10.2473/shigentosozai.116.929>.
- [30] V.T. Nguyen, K. Binnemans, S. Riaño, Refining of palladium, platinum, and rhodium from a pregnant leach solution of spent automotive catalysts by continuous counter-current solvent extraction with ionic liquids in mixer-settlers, *Ind. Eng. Chem. Res.* 62 (23) (2023) 9549–9562, <https://doi.org/10.1021/acs.iecr.3c00851>.
- [31] E. Garcíadiego-Ortega, D. Tsoulidis, M. Pineda, E.S. Fraga, P. Angeli, Hydrodynamics and mass transfer in segmented flow small channel contactors for uranium extraction, *Chem. Eng. Process.* 153 (2020) 107921, <https://doi.org/10.1016/j.cep.2020.107921>.
- [32] V. Dore, D. Tsoulidis, P. Angeli, Mixing patterns in water plugs during water/ionic liquid segmented flow in microchannels, *Chem. Eng. Sci.* 80 (2012) 1–11, <https://doi.org/10.1016/j.ces.2012.06.030>.
- [33] C.-Y. Yin, A.N. Nikoloski, M. Wang, Microfluidic solvent extraction of platinum and palladium from a chloride leach solution using Alamine 336, *Miner. Eng.* 45 (2013) 18–21, <https://doi.org/10.1016/j.mineng.2013.01.013>.
- [34] H. Aminian, C. Bazin, D. Hodouin, Residence time distributions in SX-EW equipment, *Int. J. Miner. Process.* 54 (3–4) (1998) 235–242, [https://doi.org/10.1016/S0301-7516\(98\)00010-6](https://doi.org/10.1016/S0301-7516(98)00010-6).
- [35] Sharma, M. M., Doraiswamy, L. K. (1985). Heterogeneous reactions: Analysis, examples, and reactor design. (Vol. 31). <https://doi.org/10.1002/aic.690310822>.